Python 3.15 Data Engineering Readiness Assessment

General Availability - Python 3.15.2 - tested 2026-10-27

94%

overall readiness | 15 PASS - 0 INCOMPAT - 2 BLOCKED - 0 SKIP - 0 FAIL (of 17 packages)

Executive summary

Python 3.15.2 was validated on macOS 27.0.1 ARM64 using uv. Of 17 packages probed, 15 passed and none failed outright. 2 did not pass: mlflow (BLOCKED); ray (BLOCKED).

Compared with 3.15.1 (2026-10-26), overall readiness moved from 94% to 94%.

Test environment

Release	3.15.2 (General Availability)
Python runtime	3.15.2
Python build	cpython-3.15.2-macos-arm64-none
Platform	macOS 27.0.1 ARM64
Test date	2026-10-27
Tester	Dr. Ceasar Jackson Jr.
Suite version	1.2.0
Docker image	pyarrow-dataeng:py3152
Environment consistent	yes

Package results

Package	Status	Version	Notes
numpy	**PASS**	2.5.4	
pandas	**PASS**	3.1.0	
polars	**PASS**	1.41.2	
duckdb	**PASS**	2.0.0	
sqlalchemy	**PASS**	2.0.50	
pydantic	**PASS**	2.14.1	
matplotlib	**PASS**	3.11.2	
plotly	**PASS**	6.7.0	

Package	Status	Version	Notes
jupyterlab	**PASS**	4.6.4	No persistent server session tested; the kernel logged an unencrypted-TCP-transport warning. Version differs from 3.15.1 (4.5.7).
sqlite3	**PASS**	3.54.1	
pyarrow	**PASS**	26.0.0	
dask.dataframe	**PASS**	2026.7.1	dask/config.py opens a file without encoding=, which raises EncodingWarning under -X warn_default_encoding (an error with -W error). Upstream, informational.
prefect	**PASS**	3.9.0	Prefect 3.7.x is unusable: it requires starlette>=1.0.1 but calls Body(min_items=...), which FastAPI 0.162.0 rejects, and no FastAPI satisfying starlette>=1.0.1 and accepting it resolves; 3.9.0 resolves with FastAPI 0.150.0. Some runs log a telemetry-heartbeat 'database is locked' error (absent with telemetry disabled) and intermittent EventsWorker errors. The EventsWorker errors reproduced identically on Python 3.14.10 with the same Prefect version during the 3.15.1 cycle (not rerun for 3.15.2); root cause undetermined; results were correct in every run.
mlflow	**BLOCKED**	unavailable	Upstream pin pyarrow<26 (every release from 3.15.0) plus no cp315 wheel for pyarrow 25.x; pyarrow 26.0.0, which has one, is excluded by the cap. The 3.15.1 workarounds (mlflow-skinny; full mlflow with a pyarrow>=26 override, which fails uv pip check) and the pyarrow 25.x source-build failure were not re-run on 3.15.2.
ray	**BLOCKED**	unavailable	no cp315 wheels on the configured index
pyspark	**PASS**	4.2.0	Local mode only; cluster mode not tested. Evidence is from the py3152 Docker image, not the macOS venv.
apache-airflow	**PASS**	3.3.2	dags test runs tasks in-process; scheduler and api-server not exercised. Binary-only resolution fails only because PyYAML 6.0.3 is sdist-only; the install builds it from source (libcst 1.9.0 resolved).

Changes since 3.15.1

17 package(s) unchanged.

Coverage changes (moving into or out of SKIP) are reported separately from compatibility changes: a package that was not measured before is not evidence of a regression or an improvement in Python itself.

Evidence and limitations

- pyspark was exercised inside the Docker image tagged 'py3152', not in the host virtual environment. Its recorded evidence reports Python 3.15.2 inside that image, so it is evidence for this release, with the scope limits stated in its note.
- Sensitivity analysis only: the headline 94% includes the recorded Docker-sourced results. Hypothetically excluding all of them from measurement (SKIP), regardless of whether their Python runtime was verified, would produce readiness of 94%.
- mlflow is BLOCKED: Upstream pin pyarrow<26 (every release from 3.15.0) plus no cp315 wheel for pyarrow 25.x; pyarrow 26.0.0, which has one, is excluded by the cap. The 3.15.1 workarounds

(mlflow-skinny; full mlflow with a pyarrow>=26 override, which fails uv pip check) and the pyarrow 25.x source-build failure were not re-run on 3.15.2..

- ray is BLOCKED: no cp315 wheels on the configured index.

How readiness is calculated

Readiness = (PASS + 0.50 x BLOCKED + 0.25 x INCOMPAT) / (packages - SKIP), rounded to a whole percent. SKIP is excluded because an unmeasured package is neither compatible nor incompatible. The weights are this project's convention, not an industry standard, and the score summarises this suite's probes only; it is not a certification of any package or deployment.